

Tatenda Mandiudza

Full-Stack Software Developer

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[LinkedIn](#) [Portfolio](#) [GitHub](#)

Professional Summary

Passionate software developer with a strong foundation in full-stack development and a proven track record of building robust applications from scratch. Experienced in creating custom web servers, RESTful APIs, and modern web applications using C#, ASP.NET, and React. Demonstrated expertise in clean architecture, asynchronous programming, and real-time communication systems.

Education

Bachelor of Computer Information Systems

University of the Free State

Jan 2021 – Dec 2023

Technical Skills

Backend Development: C#, ASP.NET Core, .NET 9, REST APIs, WebSockets, Middleware Architecture

Frontend Development: React, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap

Database & Data: MySQL, Entity Framework, Async Programming

Architecture & Patterns: Clean Architecture, MVC, Dependency Injection, SOLID Principles

Development Practices: Git, API Design, Authentication & Authorization, CORS, Rate Limiting

Professional Skills: Problem Solving, Team Collaboration, Communication, Time Management, Leadership

Featured Projects

Custom HTTP Web Server from Scratch

[View on GitHub](#)

C#, TCP/IP, HTTP Protocol, WebSockets

- Engineered a production-grade HTTP server entirely from scratch without frameworks, demonstrating deep understanding of networking protocols and server architecture.
- Implemented manual HTTP request parsing, response generation, and asynchronous multi-client handling using async/await patterns and thread-safe concurrent collections.
- Designed and built a flexible middleware pipeline supporting global error handling, request/response logging, performance monitoring, CORS policies, and rate limiting.
- Created RESTful API endpoints with dynamic routing system, comprehensive CRUD operations, and proper HTTP status code handling.
- Developed WebSocket protocol implementation from scratch, enabling real-time bidirectional communication for a multi-user chat application.
- Optimized for concurrent user handling with efficient resource management and scalable architecture.

E-Commerce REST API

[View on GitHub](#)

ASP.NET Core, .NET 9, Clean Architecture, Entity Framework, JWT

- Architected and developed a comprehensive RESTful API following Clean Architecture and SOLID principles for maximum maintainability and scalability.
- Implemented complete e-commerce functionality including product catalog management, shopping cart operations, order processing workflow, and inventory tracking.
- Integrated ASP.NET Core Identity with role-based authorization (Admin, Customer) and JWT token authentication for secure access control.
- Designed normalized database schema with Entity Framework Core for efficient data persistence and query optimization.
- Applied repository pattern and dependency injection for clean separation of concerns and testability.

Expense Tracker Application

[View on GitHub](#)

ASP.NET Core, MySQL, Bootstrap, MVC

- Built a full-stack financial management application with intuitive UI for tracking income and expenses.
- Implemented complete CRUD functionality with data validation and persistent MySQL database storage.
- Designed responsive interface using Bootstrap for seamless cross-device experience.

Notetify - Note Taking Application

[View on GitHub](#)

ASP.NET Core, MySQL, Responsive Design

- Developed a feature-rich note-taking web application with clean, modern UI and seamless user experience.
- Implemented secure data persistence with MySQL and efficient query handling.
- Created responsive design ensuring optimal viewing across desktop, tablet, and mobile devices.

Professional Experience

Student Assistant – Advanced Computer Literacy

Aug 2023 – Dec 2023

University of the Free State, Bloemfontein

- Provided technical guidance and mentorship to students during hands-on practical sessions.
- Troubleshoot technical issues and explained complex concepts in accessible terms.
- Fostered collaborative learning environment and supported diverse learning styles.

Student Assistant – VBA for Applications (Excel)

Aug 2022 – Dec 2022

University of the Free State, Bloemfontein

- Mentored students in VBA programming and Excel automation during practical sessions.
- Assisted in designing, managing, and evaluating practical assessments and projects.
- Demonstrated strong communication skills in explaining programming concepts to beginners.